

Isabella Cavalcanti.

Urban Policy and Data Analyst | Digital Government Specialist | GIS, Python and Parametric Design | Public Innovation and Sustainable Cities

isabellaeloy.be@gmail.com | github.com/bellaeloy | [Orcid](#)

Introduction

Urban Data Analyst and Researcher with experience at the intersection of urban planning, data science, and digital innovation. Skilled in data integration, spatial analysis, GIS, indicator development, and evidence-based decision-making for public sector and urban development projects. Experienced in working with multidisciplinary teams across universities, public institutions, and civil society organizations to transform complex datasets into actionable insights. Interested in leveraging programming, geospatial technologies, and data-driven approaches to address urban challenges, improve public policies, and support the development of smarter and more sustainable cities.

Education

University of São Paulo (USP) PhD in Architecture, Urbanism, and Technology	2026
University of Campinas (Unicamp) Specialization in Architecture, Urbanism, and Civil Engineering	2021
Federal University of Pernambuco (UFPE) Master's Degree in Urban Development	2020
Federal University of Campina Grande (UFCG) Bachelor's Degree in Architecture and Urbanism	2018

Professional experience

Researcher CnPq	2016-2020
- Conducted research involving parametric design for urban planning and architecture strategies; - Developed analytical methodologies supporting planning and policy evaluation based on Christopher Alexander's work; - Produced technical studies and evidence-based recommendations;	
Computacional Designer and project manager Arch&Type	2021
- Applied parametric design and computational tools for scenario modeling and spatial analysis in real estate development projects; - Designed spatial modeling workflows and analytical methodologies to support data-driven planning and investment decisions;	

- Collaborated with FGMF Architects on real estate studies initiatives.

Urban planer and researcher | [Unicamp](#)

2020-2021

- Performed geospatial and territorial analyses focused on sustainability, biodiversity, and urban development for the International Hub for Sustainable Development (HIDS);
- Worked closely with researchers, planners, and public-sector stakeholders, facilitating workshops and collaborative planning activities;
- Developed maps, spatial datasets, and technical reports to inform strategic planning and data-driven decision-making.

Computacional Designer and project manager | [Mookamboo Architecture](#)

2021-2022

- Developed spatial studies, site assessments, and feasibility analyses for urban planning and architectural development projects;
- Coordinated the work of junior architects and led feasibility studies, ensuring the quality and timely delivery of project outputs.

Architect, analyst and urban planner| [PExURB and USP](#)

- Produced cartographic outputs, indicators, and analytical reports for Itaipuaçu's masterplan in Rio de Janeiro State;
- Performed spatial analysis and territorial diagnostics using environmental, regulatory, and socio-economic datasets.

Data visualization specialist | [PAHO/WHO and Unicamp](#)

2022-2023

- Led the graphic design and visual production of the book *Health Promotion and Cities (2024)*;
- Produced editorial layouts, visual content, and communication materials to enhance the accessibility and dissemination of academic and technical knowledge;

Researcher | [CAPES](#)

2024-2026

- Managed, integrated, and analyzed complex datasets from multiple sources to support PhD research on urban data and governance;
- Applied programming languages and complex network analysis methods to investigate urban and socio-technical systems;
- Led study groups focused on the theory and practice of data governance, fostering interdisciplinary knowledge exchange;
- Delivered undergraduate teaching activities and facilitated workshops on remote sensing and geospatial analysis;
- Served as a member of the editorial committee of the scientific journal VIRUS for three years, contributing to the review and dissemination of

interdisciplinary research.

Skills

Data analytics and programming	Data Analysis • Data Integration • Data Management • Data Governance • Data Quality Control • Monitoring & Evaluation (M&E) • KPI Development • Quantitative & Qualitative Analysis • Python • SQL
GIS and spatial analytics	Geospatial Data Management • Spatial Modeling • Cartography • Remote Sensing • Territorial Diagnostics • Land-Use Analysis • Environmental Assessment • Geospatial Visualization
Urban planning and public policy	Evidence-Based Policy Analysis • Territorial Development • Sustainable Cities • Public Sector Innovation • Digital Government in Brazil
Data visualization and communication	Power BI • Data Visualization • Technical Reporting • Data Storytelling • Stakeholder Engagement
Software and tools	Python • SQL • QGIS • Excel • Datawrapper • Figma • Grasshopper • Rhinoceros 3D • AutoCAD • SketchUp • Lightroom • Adobe suite
Languages	Portuguese (Native) • English (Advanced) • Spanish (Basic) • French (Basic Reading)
Recognitions and Awards	• Academic Laureate: 'Professor Lynaldo Cavalcanti de Albuquerque' Award 2018 • Finalist – MIT Policy Hackathon “Environmental Justice, Time AmbientALL”, Massachusetts Institute of Technology (MIT), 2020 • 3rd Place, ABAP Roberto Burle Marx Award – Landscape Planning Practice Category 2024 • Exhibition at UFPE Museum for Banco Cabaça, developed at PPlab in partnership with Nando Albuquerque 2025